

Matrix Calculus and Optimization

Lecture Notes: From Gradients to Machine Learning

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1 Introduction to Matrix Calculus and Optimization

Course Timeline: Weeks 1-2

This lecture covers the fundamental principles of matrix calculus with applications to optimization problems. Topics include gradient computation, derivative rules, quadratic forms, and derivation of normal equations.

1.1 Historical Context and Motivation

Matrix calculus emerged from the need to extend calculus to multivariate systems. While single-variable calculus handles functions $f : \mathbb{R} \rightarrow \mathbb{R}$, modern applications in data science, machine learning, and engineering require optimization of functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Why This Matters for Math Students

For mathematics students, understanding matrix calculus is crucial because:

- **Bridging Theory and Application:** Connects abstract linear algebra with practical optimization
- **Foundation for Advanced Topics:** Essential for numerical analysis, optimization theory, and machine learning
- **Research Readiness:** Prepares for graduate studies in applied mathematics, statistics, and data science
- **Problem-Solving Skills:** Develops ability to solve high-dimensional optimization problems efficiently

2 Theoretical Foundations

2.1 Single Variable Optimization Review

For a smooth function $f : \mathbb{R} \rightarrow \mathbb{R}$, finding optima involves:

1. **First-order condition:** $f'(x) = 0$ (necessary but not sufficient)
2. **Second-order condition:**
 - Minimum: $f''(x) > 0$
 - Maximum: $f''(x) < 0$

Step-by-Step Example

Example: Find minimum of $f(x) = x^2 - 4x + 7$

$$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$$

$$f''(x) = 2 > 0 \Rightarrow \text{Minimum at } x = 2$$

2.2 Extension to Multiple Variables: The Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient replaces the derivative:

💡 Theoretical Foundations

Definition: The gradient $\nabla f(\mathbf{x})$ is a **column vector** of partial derivatives:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Geometric Interpretation: The gradient points in the direction of **steepest ascent**. For minimization, we follow $-\nabla f(\mathbf{x})$.

Optimization Condition: For unconstrained optimization, a necessary condition for $\mathbf{x}^*$ to be an optimum is:

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

3 Essential Derivative Rules with Proofs

3.1 Rule 1: Linear Functions

For $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} = \sum_{i=1}^n c_i x_i$:

$$\begin{aligned} \frac{\partial f}{\partial x_i} &= c_i \quad \text{for all } i \\ \nabla f(\mathbf{x}) &= \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \mathbf{c} \end{aligned}$$

3.2 Rule 2: Squared 2-Norm

For $f(\mathbf{x}) = \|\mathbf{x}\|^2 = \sum_{i=1}^n x_i^2$:

$$\frac{\partial}{\partial x_i} \sum_{j=1}^n x_j^2 = 2x_i \quad \Rightarrow \quad \nabla f(\mathbf{x}) = 2\mathbf{x}$$

3.3 Rule 3: Quadratic Forms

For $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$, where $Q \in \mathbb{R}^{n \times n}$:

💡 Theoretical Foundations

Derivation:

$$f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j$$

Partial derivative with respect to x_k :

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n q_{kj} x_j + \sum_{i=1}^n q_{ik} x_i$$

Thus:

$$\nabla f(\mathbf{x}) = Q\mathbf{x} + Q^T\mathbf{x} = (Q + Q^T)\mathbf{x}$$

Special Case: If Q is symmetric ($Q = Q^T$):

$$\nabla f(\mathbf{x}) = 2Q\mathbf{x}$$

Why assume symmetry? For any matrix Q :

$$\mathbf{x}^T Q \mathbf{x} = \mathbf{x}^T \left(\frac{Q + Q^T}{2} \right) \mathbf{x}$$

Only the symmetric part affects the quadratic form.

4 Derivation of Normal Equations

4.1 Problem Formulation

Given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{y} \in \mathbb{R}^m$, find $\mathbf{x} \in \mathbb{R}^n$ minimizing:

$$f(\mathbf{x}) = \|\mathbf{y} - A\mathbf{x}\|^2$$

Step-by-Step Example

Step-by-Step Derivation:

1. **Rewrite:** $f(\mathbf{x}) = (\mathbf{y} - A\mathbf{x})^T(\mathbf{y} - A\mathbf{x})$

2. **Expand:**

$$\begin{aligned} f(\mathbf{x}) &= (\mathbf{y}^T - \mathbf{x}^T A^T)(\mathbf{y} - A\mathbf{x}) \\ &= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T A\mathbf{x} - \mathbf{x}^T A^T \mathbf{y} + \mathbf{x}^T A^T A\mathbf{x} \end{aligned}$$

3. **Simplify:** Since $\mathbf{y}^T A\mathbf{x}$ is scalar:

$$\mathbf{y}^T A\mathbf{x} = (\mathbf{y}^T A\mathbf{x})^T = \mathbf{x}^T A^T \mathbf{y}$$

Thus:

$$f(\mathbf{x}) = \mathbf{x}^T (A^T A)\mathbf{x} - 2\mathbf{y}^T A\mathbf{x} + \mathbf{y}^T \mathbf{y}$$

4. **Compute Gradient:**

$$\nabla[\mathbf{x}^T (A^T A)\mathbf{x}] = 2(A^T A)\mathbf{x} \quad (\text{since } A^T A \text{ symmetric})$$

$$\nabla[-2\mathbf{y}^T A\mathbf{x}] = -2A^T \mathbf{y}$$

$$\nabla[\mathbf{y}^T \mathbf{y}] = \mathbf{0}$$

$$\nabla f(\mathbf{x}) = 2(A^T A)\mathbf{x} - 2A^T \mathbf{y}$$

5. **Set to Zero:**

$$2(A^T A)\mathbf{x} - 2A^T \mathbf{y} = \mathbf{0} \quad \Rightarrow \quad (A^T A)\mathbf{x} = A^T \mathbf{y}$$

6. **Solution** (if $A^T A$ invertible):

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{y}$$

5 Quadratic Forms and Definite Matrices

5.1 Geometric Interpretation

A quadratic form $f(\mathbf{x}) = \mathbf{x}^T Q\mathbf{x}$ represents:

- **Ellipsoid/Paraboloid** opening upward if $Q > 0$
- **Saddle surface** if Q is indefinite
- **Contour lines** determined by eigenvalues of Q

5.2 Definiteness Classification

For symmetric Q :

Type	Condition	Eigenvalues
Positive Definite ($Q > 0$)	$\mathbf{x}^T Q \mathbf{x} > 0, \forall \mathbf{x} \neq \mathbf{0}$	All > 0
Positive Semi-Definite ($Q \geq 0$)	$\mathbf{x}^T Q \mathbf{x} \geq 0, \forall \mathbf{x}$	All ≥ 0
Negative Definite ($Q < 0$)	$\mathbf{x}^T Q \mathbf{x} < 0, \forall \mathbf{x} \neq \mathbf{0}$	All < 0
Indefinite	Takes both positive and negative values	Mixed signs

Table 1: Matrix Definiteness Classification

💡 Theoretical Foundations

Why $A^T A$ is Always Positive Semi-Definite:

$$\mathbf{x}^T (A^T A) \mathbf{x} = (A\mathbf{x})^T (A\mathbf{x}) = \|A\mathbf{x}\|^2 \geq 0$$

If A has full column rank, $A\mathbf{x} \neq \mathbf{0}$ for $\mathbf{x} \neq \mathbf{0}$, making $A^T A$ positive definite.

6 Practical Applications in Data Science & ML

🏠 Practical Applications

6.1 Application 1: Linear Regression (OLS)

Problem: Given (x_i, y_i) , find β_0, β_1 minimizing $\sum (y_i - (\beta_0 + \beta_1 x_i))^2$

Matrix Form:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

Solution: $\beta = (X^T X)^{-1} X^T \mathbf{y}$

6.2 Application 2: Gradient Descent

When $A^T A$ too large:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$$

where $\nabla f(\mathbf{x}) = 2(A^T A \mathbf{x} - A^T \mathbf{y})$

6.3 Application 3: Ridge Regression

Add ℓ_2 regularization:

$$\min_{\mathbf{x}} \|\mathbf{y} - A\mathbf{x}\|^2 + \lambda \|\mathbf{x}\|^2$$

Solution: $(A^T A + \lambda I) \mathbf{x} = A^T \mathbf{y}$

6.4 Application 4: Neural Networks

Backpropagation uses chain rule with matrix derivatives:

- Loss gradients flow backward through layers
- Weight updates: $\Delta W = -\alpha \frac{\partial L}{\partial W}$
- Optimization: Adam, RMSProp build on gradient concepts

7 Step-by-Step Worked Examples

7.1 Example 1: Gradient Computation

Step-by-Step Example

Find $\nabla f(\mathbf{x})$ for $f(\mathbf{x}) = 3x_1^2 + 2x_1x_2 + 4x_2^2 - 5x_1 + 6x_2$

Solution:

1. Write in matrix form: $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x}$

$$Q = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

2. Quadratic part: $2Q\mathbf{x} = 2 \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6x_1 + 2x_2 \\ 2x_1 + 8x_2 \end{bmatrix}$

3. Linear part: $\mathbf{c} = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$

4. Combine: $\nabla f(\mathbf{x}) = \begin{bmatrix} 6x_1 + 2x_2 - 5 \\ 2x_1 + 8x_2 + 6 \end{bmatrix}$

7.2 Example 2: Minimization Problem

Step-by-Step Example

Minimize $f(x_1, x_2) = x_1^2 + 4x_1x_2 + 5x_2^2 - 2x_1 + 3x_2$

Solution:

1. Gradient: $\nabla f = \begin{bmatrix} 2x_1 + 4x_2 - 2 \\ 4x_1 + 10x_2 + 3 \end{bmatrix}$

2. Set to zero:

$$\begin{cases} 2x_1 + 4x_2 = 2 \\ 4x_1 + 10x_2 = -3 \end{cases}$$

3. Solve: $x_1 = 8, x_2 = -3.5$

4. Check Hessian: $H = \begin{bmatrix} 2 & 4 \\ 4 & 10 \end{bmatrix}$

- Determinant: $20 - 16 = 4 > 0$
- Leading principal minor: $2 > 0$
- $\Rightarrow$ Positive definite $\Rightarrow$ Minimum confirmed

8 Practice Problems

☰ Practice Problems

8.1 Problem 1: Gradient Computation

Find $\nabla f(\mathbf{x})$ for:

(a) $f(\mathbf{x}) = 2x_1^2 - 3x_1x_2 + 4x_2^2$

(b) $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} + \mathbf{x}^T B \mathbf{x}$ where $\mathbf{a} = [1, 2]^T$, $B = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

8.2 Problem 2: Critical Points Analysis

Find and classify critical points of:

$$f(x_1, x_2) = x_1^2 + 2x_2^2 + 2x_1x_2 - 4x_1 - 6x_2$$

8.3 Problem 3: Normal Equations

Given $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix}$, solve $A^T A \mathbf{x} = A^T \mathbf{y}$

8.4 Problem 4: Definiteness Classification

Classify matrices:

(a) $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

8.5 Problem 5: Ridge Regression Derivation

Derive normal equations for:

$$\min_{\mathbf{x}} \|\mathbf{y} - A\mathbf{x}\|^2 + \lambda \|\mathbf{x}\|^2$$

Show solution is $\mathbf{x} = (A^T A + \lambda I)^{-1} A^T \mathbf{y}$

9 Summary and Key Takeaways

9.1 Core Concepts

9.2 Common Pitfalls to Avoid

- **Gradient as row vector:** Always use column vector for gradient
- **Forgetting symmetry:** Use $\frac{1}{2}(Q + Q^T)$ for quadratic forms

Concept	Key Insight
Gradient	Column vector of partial derivatives; points in steepest ascent direction
Optimization Condition	$\nabla f(\mathbf{x}) = \mathbf{0}$ necessary for interior optima
Linear Function Derivative	$\nabla(\mathbf{c}^T \mathbf{x}) = \mathbf{c}$
Quadratic Form Derivative	$\nabla(\mathbf{x}^T Q \mathbf{x}) = (Q + Q^T)\mathbf{x}$ (or $2Q\mathbf{x}$ if symmetric)
Normal Equations	$A^T A \mathbf{x} = A^T \mathbf{y}$ minimizes $\ \mathbf{y} - A\mathbf{x}\ ^2$
Positive Definite Matrices	$\mathbf{x}^T Q \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$; ensures unique minimum

Table 2: Matrix Calculus Summary

- **Assuming invertibility:** Check rank of $A^T A$ before using $(A^T A)^{-1}$
- **Confusing conditions:** $\nabla f = \mathbf{0}$ necessary but not sufficient for minimum

9.3 Further Connections

- **Hessian Matrix:** Second derivatives $\nabla^2 f(\mathbf{x})$ for sufficiency conditions
- **Convexity:** If f convex, $\nabla f(\mathbf{x}^*) = \mathbf{0}$ sufficient for global minimum
- **Lagrange Multipliers:** Extension to constrained optimization
- **Matrix Cookbook:** Reference for matrix derivatives (Petersen & Pedersen)

Advanced Reading References

Advanced Reading:

- **Books:**
 1. Boyd & Vandenberghe, *Convex Optimization* (Chapters 2-5)
 2. Strang, *Linear Algebra and Learning from Data*
 3. Magnus & Neudecker, *Matrix Differential Calculus*
- **Research Papers:**
 1. "Optimization Methods for Large-Scale Machine Learning" (Bottou et al.)
 2. "Matrix Calculus in Optimization" (Minka)
- **Online Resources:**
 1. MIT OpenCourseWare: Matrix Calculus for Machine Learning
 2. Stanford CS229: Linear Algebra Review and Matrix Calculus

Learning Resources

Learning Resources:

-  3Blue1Brown: Essence of Linear Algebra
-  3Blue1Brown: Neural Networks
-  Matrix Cookbook (Petersen & Pedersen)
-  NumPy Exercises for Matrix Operations

Appendix: Solutions to Practice Problems

Problem 1 Solutions

$$(a) \nabla f = \begin{bmatrix} 4x_1 - 3x_2 \\ -3x_1 + 8x_2 \end{bmatrix}$$

$$(b) \nabla f = \mathbf{a} + 2B\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 2x_1 - x_2 \\ -x_1 + 3x_2 \end{bmatrix}$$

Problem 2 Solution

$$\text{Gradient: } \begin{bmatrix} 2x_1 + 2x_2 - 4 \\ 4x_2 + 2x_1 - 6 \end{bmatrix} = \mathbf{0}$$

$$\text{Solution: } x_1 = 1, x_2 = 1$$

$$\text{Hessian: } \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix} \text{ positive definite} \Rightarrow \text{Minimum}$$

Problem 3 Solution

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, A^T \mathbf{y} = \begin{bmatrix} 11 \\ 17 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Problem 4 Solutions

(a) Positive definite (eigenvalues $\approx 5.56, 1.44$)

(b) Positive semi-definite (rank 1, eigenvalues 6,0,0)

(c) Positive definite (eigenvalues 3,1)

Problem 5 Derivation

$$\begin{aligned} f(\mathbf{x}) &= \|\mathbf{y} - A\mathbf{x}\|^2 + \lambda \|\mathbf{x}\|^2 \\ &= \mathbf{x}^T (A^T A) \mathbf{x} - 2\mathbf{y}^T A \mathbf{x} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{x}^T \mathbf{x} \\ \nabla f &= 2(A^T A) \mathbf{x} - 2A^T \mathbf{y} + 2\lambda \mathbf{x} \\ &= 2[(A^T A + \lambda I) \mathbf{x} - A^T \mathbf{y}] = \mathbf{0} \\ &\Rightarrow (A^T A + \lambda I) \mathbf{x} = A^T \mathbf{y} \end{aligned}$$